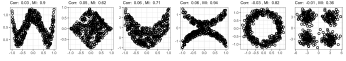
# Introduction to Machine Learning

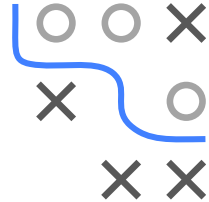
## Information Theory
## Joint Entropy and Mutual Information II



**Learning goals**

- Know mutual information as the amount of information of an RV obtained by another
- Know properties of MI

# MUTUAL INFORMATION - COROLLARIES



**Non-negativity of mutual information:** For any two random variables, $X$, $Y$, $I(X;Y) \geq 0$, with equality if and only if $X$ and $Y$ are independent.

**Proof:** $I(X;Y) = D_{KL}(p(x,y) \| p(x)p(y)) \geq 0$, with equality if and only if $p(x,y) = p(x)p(y)$ (i.e., $X$ and $Y$ are independent).

**Conditioning reduces entropy (information can't hurt):**

$$H(X|Y) \leq H(X),$$

with equality if and only if $X$ and $Y$ are independent.

**Proof:** $0 \leq I(X;Y) = H(X) - H(X|Y)$
Intuitively, the theorem says that knowing another random variable $Y$ can only reduce the uncertainty in $X$. Note that this is true only on average.

# MUTUAL INFORMATION - COROLLARIES / 2

**Independence bound on entropy:**

$$H(X_1, X_2, \ldots, X_n) \leq \sum_{i=1}^{n} H(X_i),$$

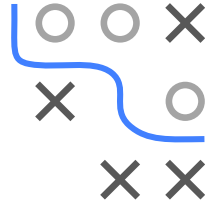
Holds with equality if and only if $X_i$ are jointly independent.

**Proof:** With chain rule and "conditioning reduces entropy"

$$H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} H(X_i | X_{i-1}, \ldots, X_1) \quad \leq \sum_{i=1}^{n} H(X_i)$$

Equality holds iff $X_i$ is independent of $X_{i-1}, \ldots, X_1$ for all $i$, so iff all $X_i$ are jointly independent.

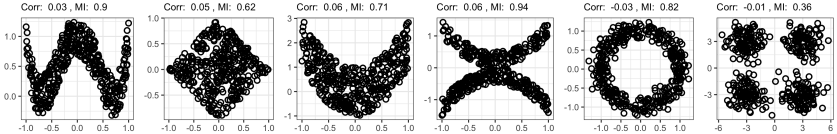
# MUTUAL INFORMATION PROPERTIES



- MI is a measure of the amount of "dependence" between variables. It is zero if and only if the variables are independent.
- OTOH, if one RV is a deterministic function of the other, MI is maximal, i.e. entropy of the first RV.
- Unlike (Pearson) correlation, MI is not limited to real-valued RVs.
- Can use MI as a **feature filter**, sometimes called information gain.
- Can also be used in CART to select feature for split.
  Splitting on MI/IG = risk reduction with log-loss.
- MI invariant under injective and continuously differentiable reparametrizations.

# MUTUAL INFORMATION VS. CORRELATION

- If two RVs are independent, their correlation is 0.
- But: two dependent RVs can have correlation 0 because correlation only measures linear dependence.




- Above: Many examples with strong dependence, nearly 0 correlation and much larger MI.
- MI can be seen as more general measure of dependence than correlation.

# MUTUAL INFORMATION - EXAMPLE

Let $X$, $Y$ be two correlated Gaussian random variables.
$(X, Y) \sim \mathcal{N}(0, K)$ with correlation $\rho$ and covariance matrix $K$:

$$K = \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}$$

Then $h(X) = h(Y) = \frac{1}{2}\log\left((2\pi e)\sigma^2\right)$, and
$h(X, Y) = \frac{1}{2}\log\left((2\pi e)^2|K|\right) = \frac{1}{2}\log\left((2\pi e)^2\sigma^4(1-\rho^2)\right)$, and thus

$$I(X; Y) = h(X) + h(Y) - h(X, Y) = -\frac{1}{2}\log(1-\rho^2).$$

For $\rho = 0$, $X$ and $Y$ are independent and $I(X; Y) = 0$.
For $\rho = \pm 1$, $X$ and $Y$ are perfectly correlated and $I(X; Y) \to \infty$.